

Exercise Adventure Works quarterly report

Introduction

In this module, you have learned about the creative processes involved in creating engaging dashboards and reports using Microsoft Power BI.

Throughout these weeks, you learned about visualization and design best practices, designing powerful report pages, and the process of data storytelling.

To create effective and engaging visualizations in Power BI desktop and service, you learned about the criteria of chart selection. This helped build towards creating and effective data story.

In a data story, the narrative created, the data selected to present this story, and the visualizations used are the key components of an effective data story.

In this exercise, you will create a story about the quarterly sales of Adventure Works in the form of a dashboard and report. This exercise will test your data visualization and storytelling skills, which are essential to convey insights effectively.

Scenario

Adventure Works, a multinational bicycle manufacturer, has multiple sales operations across the globe. The company's executive board and sales management team require quarterly sales information to assess their performance for the previous quarter. This analysis should provide them with the insights they need to make strategic decisions for their upcoming quarter.

As a data storyteller, the specific tasks you should perform in this exercise are:

1. Create a data story through a dashboard, providing an overview of quarterly sales performance for the executive board of Adventure Works.
2. Create a granular report with interactivity about the same sales period for the sales management team.
3. Highlight any trends identified, such as higher/lower sales by region, category, and salespersons.
4. Provide a conclusion summarizing the key takeaways for both the board of directors and sales management team.

By completing this exercise, you will demonstrate your ability to:

- Review the dataset and identify the measures and fields needed to support your story.
- Create a report in Power BI desktop by appropriate selection of chart which shows clarity to present your story.
- Publish your report on the Power BI service and create a dashboard for the executive board.
- Drive a few insights from your story that you can convey to stakeholders.

Step 1: Download the Power BI project file

1. Download the Power BI project file *Adventure Works Quarterly Report.pbix* by selecting the file below, and inspect the data tables and fields.
2. Download the Adventure Works logo file. You can do so by selecting the logo below and downloading it.



Step 2: Identify the relevant data fields and create visualizations

1. Identify the relevant data fields and measures that support the data story you are going to present to the sales team as well as the executive board.
2. Create visualizations by appropriate selection of charts for each data type you are presenting. For example, here is a list of chart types that you could use for different types of data:
 - a. A column/bar chart to display sales on a time scale.
 - b. A donut chart or funnel chart can show sales by category.
 - c. A map visual, a pie chart, or bar chart can be used to display sales by country.
 - d. A bar chart can be used to show the sales performance of salespeople.
 - e. A treemap can be used to visualize units sold for each category.
 - f. As the objective of the story is to show the quarterly sales performance you can add slicers for the year and quarters to interactively explore the data.

- g. You can also add top selling products, top performing regions, and salesperson to your report to support your story. Card visuals highlight key metrics for the executive board such as profit, revenue, profit margin and so on.
3. Format your visualizations to create a consistent look in terms of font size, style, and color. You need to use and customize the **Accessible City Park** theme to create a consistent look for the reports.

Step 3: Craft a data story for the sales team.

1. Add a descriptive title to your story and mention the timescale that your visualization covers. For example, what are you showing the sales team?
2. Arrange the visual elements you created in step 2 that are relevant to your story presenting to the sales management team in a logical sequence. Make sure to include the most valuable information at the top of the canvas and follow along accordingly.
3. Add an Adventure Works logo to your report.

Step 4: Craft a data story for the executive board.

1. Publish your report to **My workspace** in Power BI service. You need to make sure you apply all changes made and your report is saved to your local computer before publishing.
2. Log in to your Power BI service account and create a dashboard to tell a story about the quarterly sales performance to the executive board. Remember, you only need to pin the visuals from your report that are relevant to the executive board and support your story.
3. Add the Adventure Works logo to the dashboard.
4. Provide key takeaways from the story. You can add a text box to describe the key insights.
5. Optimize the dashboard for mobile devices.

Conclusion

With these steps, you have created a data story for the sales team and the executive board. You identified the data fields and measures relevant to each story and created visuals in Microsoft Power BI desktop. You then presented the visuals in the form of a Microsoft Power BI report and a dashboard in Power BI service to communicate valuable metrics of the business.